Research index: Mael Shawaly

research

Mathematics research by **Mael Shawaly**, in progress. Each folder is one project and contains its own README, code and results.

Project	Status	Summary
three-row-boards	Computer-assisted proof; novelty not established	Largest right-isosceles-free subset of a 3 × n board is n + 2 for n ≥ 4. Upper bound computer-assisted (potential certificate).
square-grid-a271906	Computations; no new results	Values and lower bounds for the square-grid problem (OEIS A271906). Corrected: OEIS already lists lower bounds for n = 12–15.
rectangular-boards	Partly proved; computed table	Exact two-row formula and computed values for small boards.

Status notes

- Nothing here has been peer reviewed.
- I did not find prior statements of the three-row result, but the literature search was not exhaustive. Novelty is not established.
- AI tools assisted with code, debugging, literature search and drafting. Every computational claim can be rerun from the scripts in each project. Each project says which parts are computer-assisted.